

3 Apple maggot flies are thought to have evolved from hawthorn flies by sympatric speciation.

Hawthorn flies are native to North America. They lay their eggs in the berries of the hawthorn bush.

The eggs hatch into maggots and the maggots develop into adult flies.

(a) State the meaning of the term **sympatric speciation**.

(2)

(b) Which row of the table represents sympatric speciation?

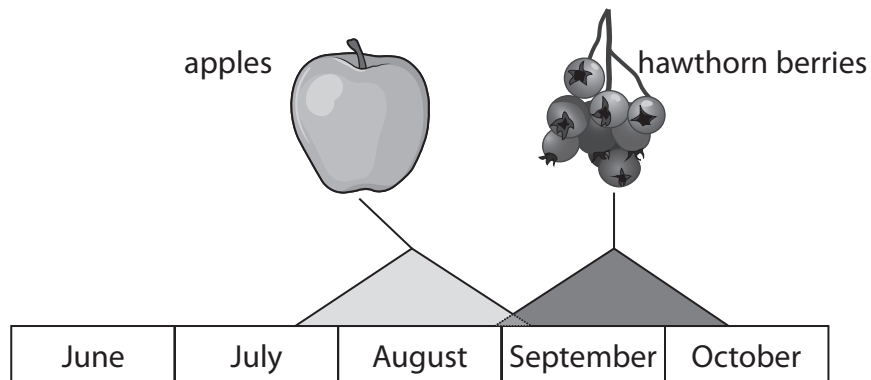
The shapes shaded dark grey represent the original species and the shapes shaded light grey represent the new species.

(1)

	Original species	First step in speciation	Final species
☐ A			
☐ B			
☐ C			
☐ D			

- (c) Following the introduction of apple trees to North America, a small group of the hawthorn flies started laying their eggs in the apples.

The diagram shows the two types of fruit and the time of year that the fruits are produced.



- (i) Suggest **two** adaptations needed for this change in behaviour of the flies.

(2)

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(ii) Explain what had to occur for changes in these flies to be inherited.

(3)

(iii) Explain why it is an advantage for the hawthorn fly and the apple maggot fly to lay their eggs in the fruits of different plants.

(2)

(iv) Suggest how these flies became reproductively isolated.

Use the information in the diagram to support your answer.

(2)

(Total for Question 3 = 12 marks)
